Probing Generative Protein Structure Models with Transition Path Sampling

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Diffusion-based structure generators sample protein conformations by reversing a stochastic denoising process, yet the stochastic landscape underlying a given prediction remains largely unexplored. Here we introduce a framework that treats the reverse diffusion of Boltz-2 as a discrete-time Markov chain on an extended state comprising coordinates, SE(3) augmentation, and per-step Gaussian noise draws, and applies transition path sampling (TPS) to collect statistically unbiased ensembles of denoising trajectories. Applied to a 228-residue two-chain co-folding task run without MSA information, 100 TPS Monte Carlo steps yield an ensemble of predicted structures with Cα RMSD spread of only 0.48–0.69 Å and a $\approx 53\%$ shooting acceptance rate, revealing that the model's stochastic landscape is sharply concentrated near a single structural attractor. The approach provides a principled diagnostic for characterizing generative model confidence, identifying structurally ambiguous predictions, and evaluating the physical quality of generated structures.

Diffusion-based structure generators such as AlphaFold3 [6], Boltz-2 [5], and RoseTTAFold All-Atom [7] are now used by millions of researchers worldwide, yet systematic evaluations reveal a consistent pattern of fragility: confidence scores are assigned to intrinsically disordered regions that cannot actually be predicted, with $\approx 18\%$ of biologically functional disordered residues hallucinated as ordered [2]; protein–ligand co-folding methods largely memorize training-set poses and fail sharply on novel compounds [3]; and antibody–antigen docking succeeds in fewer than 15% of predictions at a single seed [4]. At the scale of millions of users, a few-percent error rate is no longer a rare edge case — it is millions of structurally incorrect predictions entering drug pipelines and structural databases every year. The root diagnostic question is whether a model has genuinely learned the physical principles of folding or merely memorizes training-set attractors that score well on held-out benchmarks.

What is needed is a diagnostic tool that probes the full distribution of structures a model can produce for a fixed input, rather than accepting a single forward pass at face value. A physics-learner should yield a sharply focused, energetically reasonable ensemble for a stable fold and a diffuse one for a disordered or multi-basin sequence; a memorizer will do neither reliably. Rare-event sampling offers exactly this probe: Dorman et al. [1] recently applied transition path sampling (TPS) with Metropolis–Hastings acceptance and umbrella sampling to large language models, showing that path ensemble methods reveal failure modes entirely invisible to direct sampling. Here we take the same approach for protein structure diffusion, recasting the reverse diffusion of Boltz-2 as an explicit stochastic dynamical system and applying TPS [8, 9] to collect an unbiased ensemble of denoising trajectories connecting disordered (high-σ) to structured (low-σ) states. The key insight is that each denoising step is driven by explicit, computable noise draws, so the full path probability factorizes analytically and TPS acceptance ratios are exact.

Algorithm 1. TPS for a Generic Non-Reversible Generative Model

Require: G: forward kernel; π_i: noise prior; A, B: state volumes; $\mathcal{X}^{(0)}$: initial reactive path with $x_0 \in A$, $x_L \in B$.

for $k = 1, \ldots, N_{\mathrm{MC}}$ **do**
  Select: $i^* \sim \mathrm{Uniform}\{1, \ldots, L-1\}$
  Resample: $\xi_{i^*}^{\mathrm{new}} \sim \pi_{i^*}$; draw $\xi_i^{\mathrm{new}} \sim \pi_i$ for $i > i^*$
  Integrate: $x_{i+1}^{\mathrm{new}} = G(x_i^{\mathrm{new}}, \xi_i^{\mathrm{new}})$ for $i = i^*, \ldots, L-1$
  Reactivity: if $x_L^{\mathrm{new}} \notin B$: reject, $\mathcal{X}^{(k)} = \mathcal{X}^{(k-1)}$; continue
  Log-ratio: compute Δ via Eq. (5)
  Accept: $\mathcal{X}^{(k)} = \mathcal{X}^{\mathrm{new}}$ w.p. $\min(1, e^{\Delta})$; else $\mathcal{X}^{(k-1)}$
end for

Note. The algorithm shows *forward shooting* (suffix resampled, prefix fixed, $\Delta = 0$ always). The ergodic scheme also uses *backward shooting* (prefix via q_{bwd}, Hastings-corrected, Eq. **??**) and *global reshuffle* (fresh $x_0 \sim \rho$ and all ξ_i; always accepts, $\Delta = 0$). Jacobian $\log|\det J_i|$ is trivial for i.i.d. noise, non-trivial for EDM schedules.

TPS FOR DIFFUSION DYNAMICS

We describe the algorithm generically (Fig.), then specialize to Boltz-2. Unlike classical MD, the denoising kernel G has no time-reversal of itself, so we use a dedicated backward proposal $q_{\mathrm{bwd}}(x_i|x_{i+1})$ [15]: invert the denoiser map to recover x_i^{noisy} from x_{i+1} (rescaled fixed-point iteration [11]), draw fresh noise to propose x_i. Forward shooting (resample suffix, $\Delta = 0$), backward shooting (Δ_{bwd} corrects for $p_{\mathrm{fwd}} \neq q_{\mathrm{bwd}}$, global reshuffle (fresh x_0 and all ξ_i, always accepts) together form an ergodic move set on the reactive path ensemble [14, 15].

The Boltz-2 reverse sampler advances a coordinate tensor $x \in \mathbb{R}^{M\times 3}$ (with M padded atoms) along a fixed schedule of noise levels $\sigma_0 > \sigma_1 > \cdots > \sigma_L = 0$ over L denoising steps. We track the *extended state* $\mathcal{S}_i = (x_i, R_i, \tau_i, \varepsilon_i)$, where $R_i \in \mathrm{SO}(3)$ and $\tau_i \sim \mathcal{N}(0, s_\tau^2 I)$ are the SE(3) augmentation drawn at each step, and $\varepsilon_i \sim \mathcal{N}(0, v_i I)$ is the injected Gaussian noise with vari-

ance $v_i = \eta^2(\hat{t}^2 - \sigma_i^2)$ set by the EDM schedule [10]. One complete denoising step maps $\mathcal{S}_i \to x_{i+1}$ via

$$\tilde{x}_i = R_i\,(x_i - \bar{x}_i) + \tau_i, \tag{1}$$

$$x_i^{\text{noisy}} = \tilde{x}_i + \varepsilon_i, \tag{2}$$

$$x_{i+1} = x_i^{\text{noisy}} + s\,\frac{\sigma_i^t - \hat{t}}{\hat{t}}\,\frac{x_i^{\text{noisy}} - \hat{x}_i}{\hat{t}}, \tag{3}$$

where $\bar{x}_i$ is the mass-weighted centroid, $\hat{x}_i = c_{\text{skip}} x_i^{\text{noisy}} + c_{\text{out}} f_\theta(c_{\text{in}} x_i^{\text{noisy}}, \cdot)$ is the preconditioned denoiser output, and s is a step-scale factor. Because each step is a *deterministic* function of the random draw $(R_i, \tau_i, \varepsilon_i)$, the joint density of a full path $\mathcal{X} = \{(x_i, R_i, \tau_i, \varepsilon_i)\}_{i=0}^{L-1} \cup \{x_L\}$ factorizes as

$$\log P(\mathcal{X}) = \log \rho(x_0) + \sum_{i=0}^{L-1} \left[-\frac{\|\varepsilon_i\|^2}{2v_i} - \frac{\|\tau_i\|^2}{2s_\tau^2} + \log|\det J_i| \right], \tag{4}$$

where $\rho(x_0) \propto \mathcal{N}(0, \sigma_0^2 I)$ is the initial noise prior, and $\log|\det J_i| = 3M \log|1 + s(\sigma_i^t - \hat{t})/\hat{t}|$ is the per-step Jacobian arising from the affine map $\varepsilon_i \to x_{i+1}$ [11]. The Haar term for R_i is constant and drops from acceptance ratios. State volumes are defined on the diffusion coordinate: A at high σ (disordered) and B at low σ (structured), so every accepted path is a complete denoising trajectory.

A forward shooting move at frame i^* resamples all noise variables $(\xi_i)_{i \geq i^*}$ and forward-integrates from that point, leaving the prefix $(x_0, \ldots, x_{i^*})$ unchanged. The Metropolis–Hastings log-ratio follows directly from the factorization in Eq. (4):

$$\Delta = \sum_{i \geq i^*} \left[\log \frac{\pi_i(\xi_i^{\text{new}})}{\pi_i(\xi_i^{(k-1)})} + \log \frac{|\det J_i^{\text{new}}|}{|\det J_i^{(k-1)}|} \right]. \tag{5}$$

The proposed path is accepted with probability $\min(1, e^\Delta)$. Backward shooting at k uses the Hastings-corrected ratio $\Delta_{\text{bwd}} = [\log \pi_{\text{fwd}}(X'_{0:k}) - \log \pi_{\text{fwd}}(X_{0:k})] + [\log q_{\text{bwd}}(X_{0:k}) - \log q_{\text{bwd}}(X'_{0:k})] + \log \rho(x'_0)/\rho(x_0)$, with q_{bwd} Gaussian in the backward noise [14].

RESULTS

We apply the framework to a 228-residue two-chain heterodimer using the Boltz-2 multimer co-folding pipeline with MSA information withheld. The system comprises $M = 1792$ padded atoms, $L = 32$ denoising steps, and initial noise scale $\sigma_0 = 2560$ Å. An initial reactive trajectory is obtained from a single forward pass; TPS then runs 100 shooting MC steps using OpenPathSampling [17], with acceptance evaluated via Eq. (5).

The shooting acceptance rate is $\approx 53\%$, consistent with a well-tuned shooting distribution [9], and all accepted paths remain reactive. At every 10 MC steps, we extract the final denoised structure from the accepted path and evaluate two diagnostics: (i) Cα RMSD with respect to the initial accepted structure, and (ii) the total potential energy computed by energy minimization in OpenMM. The resulting TPS ensemble of 10 structures is shown in Fig. 1.

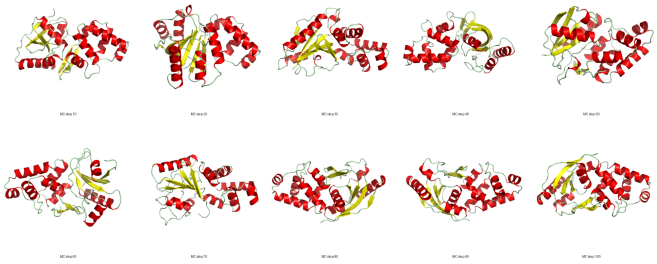

FIG. 1. **TPS-sampled structural ensemble.** Final denoised structures extracted from 10 consecutive accepted TPS paths (MC steps 10–100 in increments of 10), rendered as Cα ribbon cartoons. Both chains of the heterodimer are shown; chain coloring is preserved across panels. The overall fold and interface geometry are consistent across the ensemble, with the largest variations confined to exposed loop regions. The visual similarity is quantified in Fig. 2.

The Cα RMSD values across the ensemble span only 0.48–0.69 Å with mean 0.59 ± 0.06 Å (Fig. 2a), indicating that TPS consistently produces structures within sub-angstrom deviation of one another. The OpenMM energy ranges from $-42{,}201$ to $-41{,}653$ kJ/mol, a spread of 548 kJ/mol, with all structures converging to physically reasonable configurations (Fig. 2b). The narrow RMSD distribution is not a failure of the shooting moves to explore the stochastic space: each accepted path involves entirely different random draws $(R_i, \tau_i, \varepsilon_i)$ at every denoising step, so the structural similarity is a genuine property of the model's denoising landscape rather than a sampling artifact.

DISCUSSION

The tight RMSD distribution reflects a sharply funneled denoising landscape: despite stochastic freedom at every step, the model converges to the same structure from a wide range of intermediate noise configurations. A model uncertain about the correct fold would exhibit a broad RMSD distribution; one that is overfit would show an unnaturally narrow distribution with unphysically low energy variance. The combination of structural and energetic distributions thus provides a two-dimensional fingerprint of generative model behavior.

This perspective elevates TPS from a rare-event method into a general-purpose diagnostic. The framework applies to any generator that can be cast as an explicit Markov chain with computable transition ker-

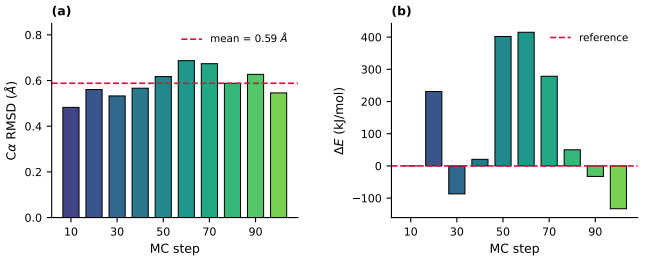

FIG. 2. **Structural and energetic distributions of the TPS ensemble.** (a) Cα RMSD values of each accepted checkpoint structure relative to the first TPS-accepted structure, for 10 structures sampled at MC steps 10–100. The distribution is narrow (0.48–0.69 Å, mean 0.59 ± 0.06 Å), indicating strong structural convergence across distinct denoising trajectories. (b) Total potential energy (kJ/mol) evaluated by OpenMM energy minimization for each TPS checkpoint structure. All structures fall in the range $-42{,}201$ to $-41{,}653$ kJ/mol with a spread of ≈ 548 kJ/mol, confirming that TPS-sampled structures are physically reasonable across the ensemble.

nels, including score-based diffusion, flow matching, and closely related architectures [12, 13]. TPS requires only that path weights are computable — satisfied here by Eq. (4) [16] — and shifts the question from "what structures does the model produce on average" to "what is the full reactive ensemble for this specific input". The full ergodic move set (forward + backward shooting + global reshuffle) ensures the Markov chain on the reactive path ensemble is irreducible regardless of the connectivity of path space [14, 15].

Several directions follow naturally. Applying TPS across MSA depths would directly quantify how evolutionary information focuses the model's stochastic landscape. For disordered or multi-basin proteins a broad TPS ensemble would serve as an early warning against over-interpreting the single-best prediction. The path weights in Eq. (4) can further be used to compute free-energy-like averages over structural order parameters without any modification of the model itself.

In summary, we show that the reverse diffusion of Boltz-2 admits a rigorous TPS formulation on an extended state that includes SE(3) augmentation and per-step noise draws, that the resulting path probability factorizes cleanly, and that 100 MC steps are sufficient to characterize the model's stochastic landscape for a 228-residue co-folding task. The sub-angstrom structural spread we find establishes that the model is highly confident in its prediction for this input, and the framework provides a direct route to interrogating generative model behavior in cases where confidence is less clear.

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